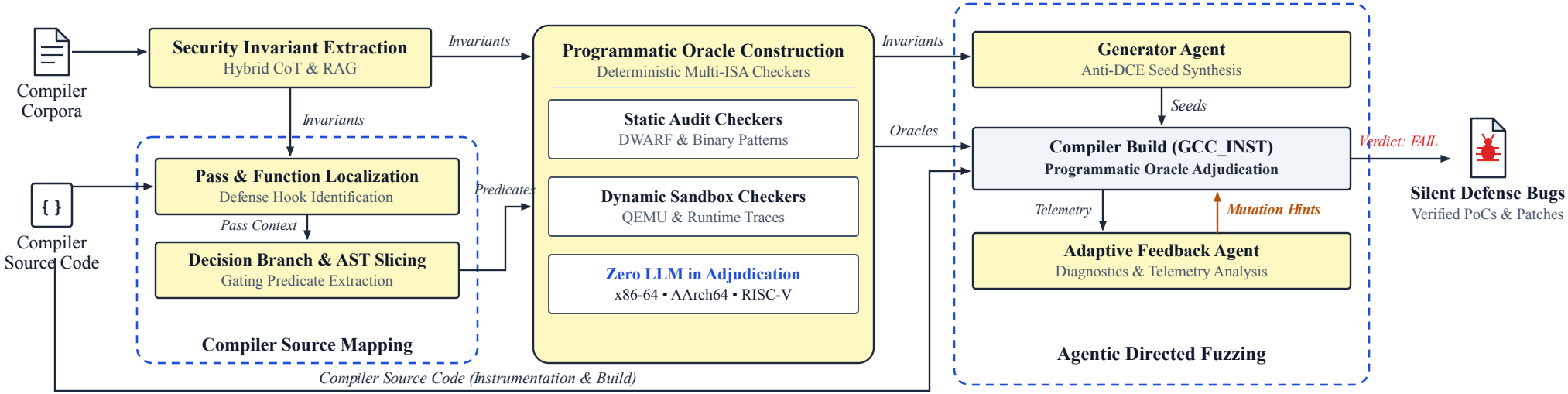

Compiler Corpora
Security Invariant Extraction
Hybrid CoT & RAG
Invariants
Programmatic Oracle Construction
Deterministic Multi-ISA Checkers
Invariants
Generator Agent
Anti-DCE Seed Synthesis
Seeds
Compiler Source Code
Pass & Function Localization
Defense Hook Identification
Pass Context
Decision Branch & AST Slicing
Gating Predicate Extraction
Predicates
Compiler Source Mapping
Static Audit Checkers
DWARF & Binary Patterns
Dynamic Sandbox Checkers
QEMU & Runtime Traces
Zero LLM in Adjudication
x86-64 • AArch64 • RISC-V
Oracles
Compiler Build (GCC_INST)
Programmatic Oracle Adjudication
Verdict: FAIL
Telemetry
Mutation Hints
Adaptive Feedback Agent
Diagnostics & Telemetry Analysis
Agentic Directed Fuzzing
Silent Defense Bugs
Verified PoCs & Patches
Compiler Source Code (Instrumentation & Build)